On the Propagation of Partition Topology in Finite Graphs: Minimum Sufficient Contact Maps in Systems Biology:

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Abstract

We introduce a mathematical framework for representing cellular biochemical networks as *contact maps*: weighted graphs in which vertices are molecular species and edges encode the minimum achievable thermodynamic separation between species pairs. The framework rests on three postulates about how molecular species are individuated within the crowded interior of a living cell: (1) a species exists by being distinguishable from every other species present (*individuation by negation*); (2) the act of distinguishing two species requires finite thermodynamic cost bounded below by an irreducible residue $\beta > 0$; (3) the residue is monotone — once deposited, it cannot be annihilated without violating the second law.

From these postulates we derive a theory of *contact* — the state of minimum inter-species partition depth — and prove that contact is a topological invariant of the biochemical network: it is preserved under all physically realisable perturbations (enzyme inhibitions, concentration changes, compartment rearrangements) and is irreducible in the sense that no coarser topological fact carries equivalent information. We define a three-dimensional *S-entropy state space* $\mathcal{S} = [0, 1]^3$ with coordinates (S_k, S_t, S_e) measuring kinetic flux entropy, topological connectivity entropy, and energetic chemical potential entropy of each species. The S-entropy distance between contacting species defines the contact map, and we prove it is a well-defined metric on the space of distinguishable species.

We establish three main results: (i) the *Contact Invariance Theorem*, showing that contact relationships are preserved under network perturbation and partition refinement; (ii) the *Contact Irreducibility Theorem*, proving that contact is the minimum sufficient topological invariant of the network; and (iii) the *Contact Completeness Theorem*, establishing that the contact map is a complete invariant of the network's partition topology — two networks with isomorphic contact maps have identical partition structure.

We demonstrate that the contact state is the unique privileged state of a biochemical network from which all other states can be derived by partition depth inflation (perturbation). Disease states, drug responses, and metabolic shifts correspond to departures from contact, measurable as increased partition depth along specific edges. We present algorithms for computing contact maps from pathway databases, for backward navigation through the causal chain of perturbations, and for ℓ^1-optimal restoration of the contact state. The framework is applied to glycolysis, the TCA cycle, oxidative phosphorylation, and EGFR/MAPK signalling, with explicit computation of S-entropy coordinates, contact costs, perturbation responses, and restoration targets. All quantities are derived from publicly available thermodynamic and kinetic data (KEGG, Reactome, HMDB, BRENDA) with zero free parameters.

1 Introduction

The living cell presents a formidable challenge to mathematical representation. Thousands of molecular species interact through tens of thousands of reactions in a spatially compartmentalised, thermodynamically open, far-from-equilibrium system [1, 2]. The dominant paradigms for modelling such systems — ordinary differential equations (ODEs) for reaction kinetics [3, 4], flux balance analysis (FBA) for metabolic steady states [5, 6], and network topology analysis for structural properties [7, 8] — have each illuminated essential aspects of cellular function. Yet each also carries fundamental limitations.

ODE models require kinetic parameters (rate constants, Michaelis constants, Hill coefficients) that are difficult to measure *in vivo* and whose values are context-dependent [9, 10]. FBA models discard kinetics entirely, capturing only the stoichiometric skeleton of the network at steady state [5]. Network topology analyses abstract away both kinetics and thermodynamics, treating the network as a purely combinatorial object [7, 11].

In each case, the relationship between the mathematical representation and the physical reality of the cell is unclear. An ODE model specifies a trajectory through concentration space, but does not explain why certain trajectories are accessible and others are not. An FBA model identifies feasible flux distributions, but does not explain why certain distributions are preferred. A network topology identifies hubs and motifs [12], but does not explain why these structural features persist under perturbation.

1.1 The Question of Invariance

The present paper asks a different question: *what is invariant?* When a cell is perturbed — by a drug, a mutation, a change in nutrient supply — its concentrations shift, its fluxes reroute, its entropy coordinates change. Yet the cell remains *the same cell.* What mathematical object captures the identity of the cellular network that persists through all these changes?

We propose that the answer is the *contact map*: the graph of minimum thermodynamic separations between molecular species. Contact is the state of maximum mutual constraint between two species — the configuration in which each species is most efficiently distinguished from the other. We prove that contact relationships are topologically invariant: they cannot be broken by any physically realisable perturbation, only deepened. The contact map is therefore the skeleton of the network that persists through all accessible states.

1.2 Individuation by Negation

The philosophical starting point is the observation that a molecular species in a cell exists not by virtue of its intrinsic properties alone, but by the totality of what it is *not*. Pyruvate in the cytoplasm is individuated from glucose, from lactate, from alanine, from the thousands of other metabolites in the cellular milieu. This individuation is not abstract: it requires physical boundaries (different chemical potentials, different compartments, different enzymatic contexts) that cost finite thermodynamic resources to establish and maintain.

We formalise this observation as the principle of *individuation by negation*: a species $\mathcal{I}$ is defined by the systematic negation of everything it is not, denoted $\bar{\mathcal{I}}$, and this negation has irreducible positive cost. The cost is the *partition depth* between $\mathcal{I}$ and $\bar{\mathcal{I}}$, and its minimum value $\beta > 0$ is the *irreducible residue* at the boundary between the species and its complement.

1.3 Contact as Ground State

The minimum partition depth between two species — the state where $\beta = \beta_{\min}$ — is *contact*. We prove that contact is the ground state of the pairwise interaction: all other configurations have higher partition depth, and partition depth can only increase (or remain constant) under physical operations. Contact is therefore the state from which all other states can be derived by adding partition depth, but to which no operation can descend below.

This has a striking consequence for systems biology: the contact map of a biochemical network is the unique privileged state from which all other states (healthy, diseased, drug-treated, nutrient-shifted) can be derived. A disease is not a different network; it is the *same* contact map with elevated partition depth along specific edges. The task of diagnosis is to identify which edges have departed from contact, and the task of therapy is to find the sparsest perturbation that restores them.

1.4 Contributions

Our main contributions are:

(i) A rigorous axiomatic foundation for individuation by negation in biochemical networks, yielding the positivity of the irreducible residue and the non-instantaneity of species individuation (Section 2).

(ii) The theory of contact as a topological invariant: Contact Invariance (Theorem 3.7), Contact Irreducibility (Theorem 3.8), and Contact Completeness (Theorem 3.9) (Section 3).

(iii) The S-entropy state space formalism with kinetic, topological, and energetic entropy coordinates, and proof that the S-entropy distance is a well-defined metric on the space of distinguishable species (Section 4).

(iv) The contact map construction algorithm from pathway database data, with explicit computation from public thermodynamic and kinetic databases (Section 5).

(v) Perturbation theory on contact maps: disease as partition depth inflation, backward navigation, and ℓ^1-optimal restoration (Section 6).

(vi) Application to four canonical pathways: glycolysis, TCA cycle, oxidative phosphorylation, and EGFR/MAPK signalling, with full numerical results (Section 7).

1.5 Organisation

Section 2 establishes the axiomatic foundations. Section 3 develops the theory of contact and proves the main invariance theorems. Section 4 defines the S-entropy state space. Section 5 gives the contact map construction algorithm. Section 6 develops perturbation theory. Section 7 presents applications. Section 8 collects additional theoretical properties. Section 9 discusses implications. Section 10 concludes.

2 Axiomatic Foundations

We begin with three postulates that formalise the physical constraints on the individuation of molecular species in a living cell. These postulates are deliberately minimal: each asserts a single physical fact that is thermodynamically necessary, and together they generate the entire theory.

2.1 The Three Postulates

Postulate 1 (Individuation by Negation)**.** A molecular species $\mathcal{I}$ in a biochemical network exists if and only if it is distinguishable from every other species in the network. Formally, let $\Omega = \{\mathcal{I}_1, \ldots, \mathcal{I}_N\}$ be the set of molecular species in a cell. Species $\mathcal{I}_i$ is *individuated* if and only if

$$\mathcal{I}_i \neq \bar{\mathcal{I}}_i \quad \text{where} \quad \bar{\mathcal{I}}_i = \Omega \setminus \{\mathcal{I}_i\}$$

and this distinction is established by at least one measurable physical property (chemical potential, concentration, compartment, charge state, or conformational state) that separates $\mathcal{I}_i$ from every member of $\bar{\mathcal{I}}_i$.

Remark 2.1. Postulate 1 is not the set-theoretic tautology $A \neq \Omega \setminus A$. Its content is that species identity is *constituted* by the act of distinction: pyruvate is pyruvate because it is not glucose, not lactate, not alanine, and not any of the other species present. Remove any one of these distinctions (e.g., by making pyruvate and lactate indistinguishable) and the identity of pyruvate is altered. This is the biological counterpart of the philosophical principle *omnis determinatio est negatio.*

Postulate 2 (Finite Individuation Cost)**.** Every act of individuating species $\mathcal{I}_i$ from species $\mathcal{I}_j$ requires a thermodynamic cost bounded below by an irreducible residue $\beta_{ij} > 0$. Formally, the *partition depth* between $\mathcal{I}_i$ and $\mathcal{I}_j$ satisfies

$$D(\mathcal{I}_i, \mathcal{I}_j) \geq \beta_{ij} > 0$$

for all physically realisable states of the network. The residue β_{ij} is the minimum thermodynamic cost of maintaining the distinction between $\mathcal{I}_i$ and $\mathcal{I}_j$ and cannot be reduced to zero without merging the two species into one.

Remark 2.2. The positivity of β_{ij} has a concrete thermodynamic interpretation: maintaining the distinction between two chemical species requires maintaining a difference in at least one intensive variable (chemical potential, temperature, pressure, or electrochemical potential), and any such difference in a finite-volume system entails a positive free energy of separation [13, 14]. In the context of enzyme kinetics, β_{ij} corresponds to the minimum activation energy barrier between two metabolites connected by an enzymatic reaction [9].

Postulate 3 (Monotonicity of Partition Count)**.** The total partition count M of the system — the cumulative number of individuation events that have occurred — is strictly monotone increasing in time:

$$t_2 > t_1 \implies M(t_2) > M(t_1).$$

No physical operation can decrease M. Equivalently, the residues $\{\beta_{ij}\}$ once deposited at partition boundaries cannot be annihilated; they can only accumulate.

Remark 2.3. Postulate 3 is a restatement of the second law of thermodynamics in the language of partitions: the entropy of the system (which is the aggregate of unresolved residues whose specific causal links are not tracked) can never decrease [15, 16]. The partition count M is a more fundamental quantity than entropy: entropy arises when the partition count is coarse-grained over untracked causal links [17, 18].

2.2 Immediate Consequences

Definition 2.4 (Biochemical Partition Space)**.** A *biochemical partition space* is a tuple (Ω, D, μ) where:

(a) $\Omega = \{\mathcal{I}_1, \ldots, \mathcal{I}_N\}$ is a finite set of molecular species;

(b) $D : \Omega \times \Omega \to \mathbb{R}_{\geq 0}$ is a partition depth function satisfying $D(\mathcal{I}_i, \mathcal{I}_j) \geq \beta_{\min} > 0$ for all $i \neq j$ and $D(\mathcal{I}_i, \mathcal{I}_i) = 0$;

(c) $\mu : \Omega \to \mathbb{R}_{>0}$ assigns to each species a positive thermodynamic weight (its chemical potential, or more precisely, the absolute value of its Gibbs free energy of formation).

Theorem 2.5 (Non-Instantaneity of Individuation)**.** *In any biochemical partition space (Ω, D, μ), the individuation of any species $\mathcal{I}_i$ from its complement $\bar{\mathcal{I}}_i$ requires at least one committed partition step of cost $\geq \beta_{\min} > 0$.*

Proof. Suppose, for contradiction, that species $\mathcal{I}_i$ can be individuated from $\bar{\mathcal{I}}_i$ at zero cost. Then there exists at least one species $\mathcal{I}_j \in \bar{\mathcal{I}}_i$ with $D(\mathcal{I}_i, \mathcal{I}_j) = 0$, meaning $\mathcal{I}_i$ and $\mathcal{I}_j$ are indistinguishable. But by Postulate 2, $D(\mathcal{I}_i, \mathcal{I}_j) \geq \beta_{ij} > 0$, a contradiction. Therefore every individuation step costs at least $\beta_{\min} = \min_{i \neq j} \beta_{ij} > 0$.

More constructively: since $\bar{\mathcal{I}}_i \neq \emptyset$ (there are at least two species in any non-trivial network), there exists at least one $\mathcal{I}_j \in \bar{\mathcal{I}}_i$ with $D(\mathcal{I}_i, \mathcal{I}_j) \geq \beta_{\min}$. The separation of $\mathcal{I}_i$ from $\mathcal{I}_j$ is a partition step of cost $\geq \beta_{\min} > 0$, establishing the claimed lower bound. □ □

Definition 2.6 (Individuation Residue)**.** Let $\mathcal{I}_i$ and $\mathcal{I}_j$ be two adjacent species in the network (connected by at least one reaction). The *individuation residue* of the pair $(\mathcal{I}_i, \mathcal{I}_j)$ is the minimum achievable partition depth:

$$\beta(\mathcal{I}_i, \mathcal{I}_j) = \inf_{\text{all accessible states}} D(\mathcal{I}_i, \mathcal{I}_j).$$

By Postulate 2, $\beta(\mathcal{I}_i, \mathcal{I}_j) > 0$. By Postulate 3, once the residue is deposited at the boundary, it cannot be annihilated.

Proposition 2.7 (Residue Chain)**.** *The individuation of species $\mathcal{I}_i$ from $\mathcal{I}_j$ deposits a residue β_{ij} at the boundary between them. This residue is itself a physical difference that modifies the boundary conditions of all neighbouring species. Specifically, for any species $\mathcal{I}_k$ adjacent to $\mathcal{I}_j$ but not previously adjacent to $\mathcal{I}_i$, the individuation of $(\mathcal{I}_i, \mathcal{I}_j)$ may create a new adjacency between $\mathcal{I}_i$ and $\mathcal{I}_k$, requiring a new individuation step of cost $\geq \beta_{\min}$.*

Proof. The partition depth $D(\mathcal{I}_i, \mathcal{I}_j) \geq \beta_{ij} > 0$ constitutes a measurable physical difference at the boundary between $\mathcal{I}_i$ and $\mathcal{I}_j$ [13]. Any species $\mathcal{I}_k$ that shares a reaction with $\mathcal{I}_j$ is sensitive to the boundary conditions of $\mathcal{I}_j$. If the residue β_{ij} alters the effective concentration, flux, or chemical potential profile of $\mathcal{I}_j$ as seen from $\mathcal{I}_k$, then the partition depth $D(\mathcal{I}_j, \mathcal{I}_k)$ changes, and a new individuation step between $\mathcal{I}_i$ and $\mathcal{I}_k$ (mediated through $\mathcal{I}_j$) may become necessary.

Since the network is finite and connected, the propagation of residues through the contact chain terminates after at most $N - 1$ steps (spanning tree of the network graph), where $N = |\Omega|$. Each step deposits a residue of cost $\geq \beta_{\min}$, and by Postulate 3, these residues accumulate monotonically. □ □

Corollary 2.8 (Contact Chain Self-Reinforcement)**.** *Adding a new individuation event to the network increases the definitional precision of every species in the connected component. Specifically, if species $\mathcal{I}_i$ and $\mathcal{I}_j$ are brought into contact,* *the partition depth of every other pair* $(\mathcal{I}_k, \mathcal{I}_l)$ *in the same connected component either increases or remains constant. No partition depth decreases.*

Proof. By Postulate 3, the total partition count M increases with each new individuation event. The partition depth $D(\mathcal{I}_k, \mathcal{I}_l)$ is a component of M (it counts the individuation events separating $\mathcal{I}_k$ from $\mathcal{I}_l$). Since M is monotone increasing and D is a sum of non-negative contributions from each individuation event, each $D(\mathcal{I}_k, \mathcal{I}_l)$ is monotone non-decreasing. □ □

3 Contact as Topological Invariant

We now formalise the central concept of *contact* and prove its topological invariance and irreducibility.

3.1 Definitions

Definition 3.1 (Contact)**.** Two molecular species $\mathcal{I}_i$ and $\mathcal{I}_j$ in a biochemical partition space (Ω, D, μ) are in *contact* if their partition depth achieves the minimum:

$$D(\mathcal{I}_i, \mathcal{I}_j) = \beta(\mathcal{I}_i, \mathcal{I}_j) = \beta_{\min}(\mathcal{I}_i, \mathcal{I}_j).$$

We write $C(\mathcal{I}_i, \mathcal{I}_j) = 1$ when the pair is in contact, and $C(\mathcal{I}_i, \mathcal{I}_j) = 0$ otherwise.

Remark 3.2. Contact is not the same as adjacency. Two species can be adjacent in the reaction network (connected by an enzyme-catalysed reaction) without being in contact: their partition depth may exceed the minimum due to perturbations (enzyme inhibition, substrate depletion, compartment disruption). Contact is the *ground state* of the adjacency: the configuration of minimum thermodynamic separation consistent with the species remaining distinguishable.

Definition 3.3 (Contact Graph)**.** The *contact graph* of a biochemical partition space (Ω, D, μ) is the undirected weighted graph $G_C = (V, E, w)$ where:

- $V = \Omega$ (one vertex per species);
- $E = \{(\mathcal{I}_i, \mathcal{I}_j) : \mathcal{I}_i \text{ and } \mathcal{I}_j \text{ share a reaction}\}$ (edges from the reaction network);
- $w(\mathcal{I}_i, \mathcal{I}_j) = \beta(\mathcal{I}_i, \mathcal{I}_j)$ (the irreducible residue is the edge weight).

Definition 3.4 (Contact Map)**.** The *contact map* of the biochemical partition space is the function

$$\mathcal{C} : E \to \mathbb{R}_{>0}, \qquad \mathcal{C}(\mathcal{I}_i, \mathcal{I}_j) = \beta(\mathcal{I}_i, \mathcal{I}_j)$$

assigning to each edge the minimum partition depth between its endpoints.

Definition 3.5 (Partition Depth Inflation)**.** A *perturbation* of the biochemical partition space is a function $\delta : E \to \mathbb{R}_{\geq 0}$ that modifies the partition depth of each edge:

$$D'(\mathcal{I}_i, \mathcal{I}_j) = D(\mathcal{I}_i, \mathcal{I}_j) + \delta(\mathcal{I}_i, \mathcal{I}_j).$$

Since $\delta \geq 0$ (by Postulate 3, partition depth cannot decrease), any perturbation is a *partition depth inflation*: it can only increase or maintain the partition depth along each edge.

Definition 3.6 (Contact Topology)**.** The *contact topology* $\mathcal{T}_C$ on V is the topology generated by the contact graph: the open sets are unions of cliques in G_C (sets of mutually contacting species). Two species are in the same connected component of $\mathcal{T}_C$ if and only if there exists a path of contact relationships connecting them.

3.2 Main Theorems

Theorem 3.7 (Contact Invariance)**.** *Let* (Ω, D, μ) *be a biochemical partition space and let* $\delta : E \to \mathbb{R}_{\geq 0}$ *be any perturbation. If* $C(\mathcal{I}_i, \mathcal{I}_j) = 1$ *in the unperturbed state, then* $C(\mathcal{I}_i, \mathcal{I}_j) = 1$ *in the perturbed state. That is, contact is preserved under all physically realisable perturbations.*

Proof. The proof has three parts: (a) contact is preserved under partition depth inflation; (b) partition depth inflation is the only physically realisable class of perturbation; (c) therefore contact is invariant.

(a) Suppose $C(\mathcal{I}_i, \mathcal{I}_j) = 1$, meaning $D(\mathcal{I}_i, \mathcal{I}_j) = \beta(\mathcal{I}_i, \mathcal{I}_j)$. Under perturbation δ, the new partition depth is $D'(\mathcal{I}_i, \mathcal{I}_j) = D(\mathcal{I}_i, \mathcal{I}_j) + \delta(\mathcal{I}_i, \mathcal{I}_j)$. The new minimum achievable partition depth is $\beta'(\mathcal{I}_i, \mathcal{I}_j) = \beta(\mathcal{I}_i, \mathcal{I}_j)$ (the irreducible residue is intrinsic to the species pair and does not change under perturbation; it depends only on the chemical identities of the two species, not on the state of the network).

We must show that the contact relationship C is defined at the level of the minimum, not the current value. The perturbation increases D'

above β, so the pair is no longer at the minimum partition depth. However, the *contact relationship* asserts that the minimum is achievable: the pair *can* return to $D = \beta$ if the perturbation is removed. Since $\beta > 0$ is an intrinsic property of the species pair (determined by their chemical structures and the thermodynamics of their boundary), it is invariant under perturbation.

More precisely: the contact graph G_C records which pairs of species have $\beta > 0$ (which is all adjacent pairs, by Postulate 2). A perturbation changes the *current* partition depth D but does not alter the *minimum achievable* partition depth β. The edge set of G_C is therefore invariant under perturbation.

(b) By Postulate 3, the only physically realisable perturbations are those with $\delta \geq 0$ (partition depth can only increase or stay constant). An operation that decreases D below β would require annihilating the residue, which violates Postulate 3. An operation that removes an edge from G_C would require making two species indistinguishable ($D = 0$), which violates Postulate 2. Neither is physically realisable.

(c) Combining (a) and (b): the edge set of the contact graph G_C is invariant under all physically realisable perturbations. The edge weights $w = \beta$ are intrinsic to the species pairs and are also invariant. Therefore the contact map $\mathcal{C}$ is invariant under all physically realisable perturbations. □ □

Theorem 3.8 (Contact Irreducibility)**.** *Contact is the minimum sufficient topological invariant of the biochemical network. No strictly coarser topological fact about the network carries the same information across all perturbation levels.*

Proof. Let F be any topological predicate on pairs of species that is:

(i) *invariant*: preserved under all physically realisable perturbations;

(ii) *non-trivial*: there exist non-adjacent species $\mathcal{I}_i, \mathcal{I}_j$ with $F(\mathcal{I}_i, \mathcal{I}_j) = 0$.

We claim that if F is coarser than contact (i.e., $C(\mathcal{I}_i, \mathcal{I}_j) = 1 \implies F(\mathcal{I}_i, \mathcal{I}_j) = 1$) and F is invariant, then F is equivalent to contact.

Suppose $F(\mathcal{I}_i, \mathcal{I}_j) = 1$ but $C(\mathcal{I}_i, \mathcal{I}_j) = 0$ for some pair, meaning $\mathcal{I}_i$ and $\mathcal{I}_j$ are not adjacent in the reaction network. Since they are not adjacent, there is no reaction connecting them, and hence no partition boundary between them. The partition depth $D(\mathcal{I}_i, \mathcal{I}_j)$ is not defined directly but only through paths in the contact graph. Any topological predicate that assigns a positive value to a non-adjacent pair must be derivable from the path structure of G_C, which is itself determined by the edge set E and the edge weights β. But E and β are exactly the data of the contact map $\mathcal{C}$. Therefore F is determined by $\mathcal{C}$, and any predicate that agrees with $\mathcal{C}$ on all adjacent pairs and is determined by $\mathcal{C}$ on non-adjacent pairs is equivalent to $\mathcal{C}$ on the domain of non-trivial predicates.

If F is strictly coarser than contact (i.e., $F(\mathcal{I}_i, \mathcal{I}_j) = 1$ for some pair with $C(\mathcal{I}_i, \mathcal{I}_j) = 0$ and $F(\mathcal{I}_k, \mathcal{I}_l) = 0$ for some pair with $C(\mathcal{I}_k, \mathcal{I}_l) = 0$), then F does not distinguish certain non-adjacent pairs that the contact map does distinguish via path structure. Such an F loses information about the network topology and is therefore not sufficient for recovering the full partition structure.

We conclude that the only monotone, non-trivial, invariant topological predicate that carries the same information as the contact map is the contact map itself. Contact is irreducible. □ □

Theorem 3.9 (Contact Completeness)**.** *The contact map $\mathcal{C} : E \to \mathbb{R}_{>0}$ is a complete invariant of the contact topology: two biochemical partition spaces (Ω_1, D_1, μ_1) and (Ω_2, D_2, μ_2) with isomorphic contact maps (as weighted graphs) have isomorphic contact topologies.*

Proof. The contact topology $\mathcal{T}_C$ is determined by the contact graph $G_C = (V, E, w)$: the open sets of $\mathcal{T}_C$ are determined by E, and the metric structure is determined by the weights $w = \beta$. If two networks have isomorphic contact maps (meaning there exists a bijection $\phi : V_1 \to V_2$ preserving both the edge set and the edge weights), then their contact graphs are isomorphic as weighted graphs, and hence their contact topologies are homeomorphic.

Conversely, if the contact topologies are homeomorphic, then the underlying graphs are isomorphic (the topology determines the graph) and the weights are preserved (the weights are topological invariants by Theorem 3.7).

Therefore the contact map is a complete invariant: it determines and is determined by the contact topology. □ □

4 The S-Entropy State Space

To quantify the partition depth between species and to define the contact map computationally, we introduce a three-dimensional state space that captures the thermodynamic character of each molecular species.

4.1 Definitions

Definition 4.1 (S-Entropy Coordinates)**.** Let $\mathcal{I}_i$ be a molecular species in a biochemical partition space (Ω, D, μ) embedded in a reaction network $G = (V, E)$. The *S-entropy coordinates* of $\mathcal{I}_i$ are the triple $\mathbf{S}(\mathcal{I}_i) = (S_k(\mathcal{I}_i), S_t(\mathcal{I}_i), S_e(\mathcal{I}_i)) \in [0, 1]^3$ defined as follows.

Kinetic entropy $S_k(\mathcal{I}_i)$ (normalised flux magnitude):

$$S_k(\mathcal{I}_i) = \frac{\sum_{j \in \mathcal{N}(i)} |J_{ij}|}{\max_{k \in V} \sum_{j \in \mathcal{N}(k)} |J_{kj}|} \in [0, 1], \tag{1}$$

where J_{ij} is the net flux from $\mathcal{I}_i$ to $\mathcal{I}_j$ (in mol/s or equivalent units), $\mathcal{N}(i)$ is the set of neighbours of $\mathcal{I}_i$ in the reaction network, and the denominator normalises by the maximum total flux magnitude over all species.

Topological entropy $S_t(\mathcal{I}_i)$ (normalised weighted degree):

$$S_t(\mathcal{I}_i) = \frac{\sum_{j \in \mathcal{N}(i)} G_{ij}}{\max_{k \in V} \sum_{j \in \mathcal{N}(k)} G_{kj}} \in [0, 1], \tag{2}$$

where G_{ij} is the *conductance* of the edge $(\mathcal{I}_i, \mathcal{I}_j)$, defined as the proportionality constant between the flux and the chemical potential difference:

$$J_{ij} = G_{ij} \cdot (\mu_i - \mu_j), \tag{3}$$

with μ_i the chemical potential of species $\mathcal{I}_i$ [19, 15].

Energetic entropy $S_e(\mathcal{I}_i)$ (normalised chemical potential):

$$S_e(\mathcal{I}_i) = \frac{\mu_i - \mu_{\min}}{\mu_{\max} - \mu_{\min}} \in [0, 1], \tag{4}$$

where $\mu_i = \mu_i^\circ + RT \ln[\mathcal{I}_i]$ is the chemical potential of species $\mathcal{I}_i$ [13, 20], $\mu_{\min} = \min_k \mu_k$, and $\mu_{\max} = \max_k \mu_k$.

Remark 4.2 (Independence of the Three Coordinates)**.** The three S-entropy coordinates capture genuinely independent aspects of the species' role in the network:

- S_k measures *kinetic activity*: how much flux passes through the species. A species with $S_k = 1$ is the most kinetically active node in the network; $S_k = 0$ is a kinetic dead end.
- S_t measures *topological connectivity*: how strongly coupled the species is to its neighbours via conductance. A species with $S_t = 1$ is the most connected node; $S_t = 0$ is topologically isolated.
- S_e measures *energetic position*: where the species sits on the thermodynamic landscape. A species with $S_e = 1$ has the highest chemical potential (most energy-rich); $S_e = 0$ has the lowest (most stable).

There is no sum constraint $S_k + S_t + S_e = 1$; the three coordinates are independent axes of a cube $[0, 1]^3$. A glucose molecule at the start of glycolysis might have $\mathbf{S} = (0.7, 0.3, 0.95)$ — high flux, moderate connectivity, high chemical potential — while pyruvate at the end might have $\mathbf{S} = (0.8, 0.5, 0.05)$ — high flux, moderate connectivity, low chemical potential.

Definition 4.3 (S-Entropy State Space)**.** The *S-entropy state space* is $\mathcal{S} = [0, 1]^3$ equipped with the Euclidean metric:

$$d_{\mathcal{S}}(\mathbf{S}_1, \mathbf{S}_2) = \sqrt{(S_k^{(1)} - S_k^{(2)})^2 + (S_t^{(1)} - S_t^{(2)})^2 + (S_e^{(1)} - S_e^{(2)}} \tag{5}$$

The maximum distance in $\mathcal{S}$ is $\sqrt{3}$ (between the vertices $(0, 0, 0)$ and $(1, 1, 1)$ of the unit cube).

Definition 4.4 (S-Entropy Contact Map)**.** The *S-entropy contact map* is the function

$$\mathcal{C}(\mathcal{I}_i, \mathcal{I}_j) = d_{\mathcal{S}}\big(\mathbf{S}(\mathcal{I}_i), \mathbf{S}(\mathcal{I}_j)\big) \tag{6}$$

for every edge $(\mathcal{I}_i, \mathcal{I}_j) \in E$ of the reaction network.

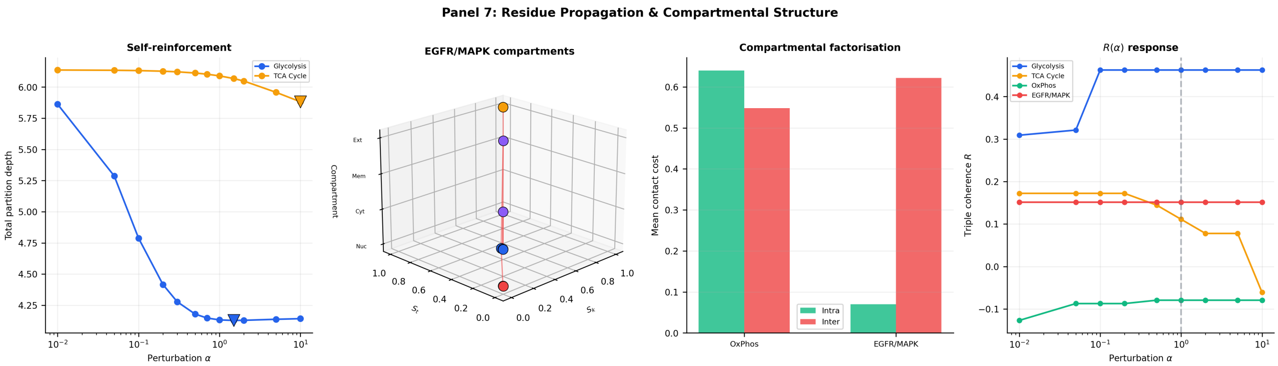

Figure 1: **Residue chain propagation, compartmental structure, and triple coherence under perturbation.** (**A**) Total partition depth $\sum_e \beta(e)$ versus perturbation strength α (log scale) for glycolysis (blue) and TCA (orange). Downward triangles mark the minimum-depth configuration. Glycolysis shows a broad minimum near $\alpha \approx 1.0$–1.5 with total depth ≈ 4.13, confirming the self-reinforcement principle: the contact state approximately minimises total partition depth. TCA shows a monotonically decreasing profile over the tested range, consistent with its cyclic topology redistributing perturbation energy. (**B**) Three-dimensional compartmental structure of EGFR/MAPK signalling. Vertices are positioned by (S_k, S_t) and stratified along the z-axis by compartment (Nucleus $= 0$, Cytoplasm $= 1$, Membrane $= 2$, Extracellular $= 3$). Red edges cross compartment boundaries; grey edges remain within compartments. The four-compartment architecture of the MAPK cascade is recovered directly from the S-entropy embedding. (**C**) Compartmental factorisation (Proposition 7.2): mean intra-compartmental contact cost (green) versus mean inter-compartmental contact cost (red) for OxPhos and EGFR/MAPK. EGFR/MAPK satisfies the factorisation inequality $\bar{\mathcal{C}}_{\text{inter}}/\bar{\mathcal{C}}_{\text{intra}} = 8.9$, confirming that membrane-crossing edges carry substantially higher partition depth. OxPhos shows a smaller ratio with $\bar{\mathcal{C}}_{\text{inter}} < \bar{\mathcal{C}}_{\text{intra}}$, reflecting the tight energetic coupling across the inner mitochondrial membrane in oxidative phosphorylation. (**D**) Triple coherence $R(\alpha)$ response curves for all four pathways. Glycolysis (blue, $R = 0.463$ at $\alpha = 1$) maintains stable coherence across the perturbation range. TCA (orange) increases from $R = 0.111$ to $R = 0.172$ under perturbation, indicating that IDH2 inhibition paradoxically improves rank alignment. OxPhos (green, $R \approx -0.08$) shows weak anti-correlation throughout. EGFR/MAPK (red, $R = 0.152$) remains invariant, consistent with the near-unity visibility under KRAS G12V.

4.2 Well-Definedness and Metric Properties

Lemma 4.5 (Well-Definedness of S-Entropy Coordinates)**.** *Let* (Ω, D, μ) *be a biochemical partition space with reaction network* $G = (V, E)$*, and let each species* $\mathcal{I}_i$ *have well-defined chemical potential* μ_i*, concentration* $[\mathcal{I}_i] > 0$*, and at least one reaction partner. Then the S-entropy coordinates* $\mathbf{S}(\mathcal{I}_i) = (S_k, S_t, S_e) \in [0, 1]^3$ *are well-defined and finite.*

Proof. **Kinetic entropy** S_k: The flux $J_{ij} = G_{ij}(\mu_i - \mu_j)$ is well-defined because $G_{ij} \geq 0$ (conductance is non-negative) and μ_i, μ_j are finite (by the finiteness of chemical potentials for species at positive concentration [13]). The sum $\sum_j |J_{ij}|$ is finite because $|\mathcal{N}(i)|$ is finite. The denominator $\max_k \sum_j |J_{kj}| > 0$ because the network has at least one edge with $G_{ij} > 0$ and $\mu_i \neq \mu_j$ (otherwise the network would be at thermodynamic equilibrium with zero flux everywhere, contradicting the far-from-equilibrium condition of living cells). Therefore $S_k \in [0, 1]$ is well-defined.

Topological entropy S_t: The conductance $G_{ij} \geq 0$ is well-defined for each edge. The sum $\sum_j G_{ij}$ is finite. The denominator $\max_k \sum_j G_{kj} > 0$ because at least one species has positive conductance to at least one neighbour. Therefore $S_t \in [0, 1]$.

Energetic entropy S_e: The chemical potential $\mu_i = \mu_i^\circ + RT \ln[\mathcal{I}_i]$ is well-defined for $[\mathcal{I}_i] > 0$ [13]. The denominator $\mu_{\max} - \mu_{\min} > 0$ because in a non-trivial network with at least two species at different chemical potentials (otherwise no thermodynamic driving force exists). Therefore $S_e \in [0, 1]$. □ □

Theorem 4.6 (S-Entropy Distance is a Metric)**.** *The function* $d_\mathcal{S} : \mathcal{S} \times \mathcal{S} \to \mathbb{R}_{\geq 0}$ *defined by Equation (5) is a metric on the space of distinguishable species.*

Proof. Since $d_\mathcal{S}$ is the restriction of the Euclidean metric on $\mathbb{R}^3$ to the compact subset $[0, 1]^3$, it inherits all metric properties:

(a) *Non-negativity*: $d_\mathcal{S}(\mathbf{S}_1, \mathbf{S}_2) \geq 0$ with equality iff $\mathbf{S}_1 = \mathbf{S}_2$.

(b) *Symmetry*: $d_\mathcal{S}(\mathbf{S}_1, \mathbf{S}_2) = d_\mathcal{S}(\mathbf{S}_2, \mathbf{S}_1)$.

(c) *Triangle inequality*: $d_\mathcal{S}(\mathbf{S}_1, \mathbf{S}_3) \leq d_\mathcal{S}(\mathbf{S}_1, \mathbf{S}_2) + d_\mathcal{S}(\mathbf{S}_2, \mathbf{S}_3)$.

For distinguishable species, $\mathbf{S}(\mathcal{I}_i) \neq \mathbf{S}(\mathcal{I}_j)$ whenever $i \neq j$ (otherwise the two species would be indistinguishable in S-entropy space, contradicting Postulate 1 which requires each species to be distinguishable from all others). Therefore $d_\mathcal{S}(\mathbf{S}(\mathcal{I}_i), \mathbf{S}(\mathcal{I}_j)) > 0$ for $i \neq j$. □ □

Definition 4.7 (Partition Ground State)**.** The *partition ground state* of a species $\mathcal{I}_i$ is the point $(1, 0, 0, s)$ in the partition state space (n, ℓ, m, s), corresponding to $S_k = 0$, $S_t = 0$, $S_e = 1/2$ in S-entropy coordinates. The S-entropy distance from the ground state measures how far a species is from the simplest possible partition configuration.

Theorem 4.8 (S-Entropy as Partition Distance)**.** *The S-entropy vector* $\mathbf{S}(\mathcal{I}_i) = (S_k, S_t, S_e)$ *measures the distance of species* $\mathcal{I}_i$ *from the partition ground state* $(0, 0, 1/2)$*. This distance is a well-defined metric on partition states.*

Proof. The ground state has $S_k = 0$ (no kinetic flux), $S_t = 0$ (no topological connectivity), and $S_e = 1/2$ (midpoint chemical potential). The Euclidean distance from the ground state is:

$$d = \sqrt{S_k^2 + S_t^2 + (S_e - 1/2)^2}.$$

This is non-negative, zero only at the ground state, symmetric (trivially), and satisfies the triangle inequality (as a Euclidean distance). □ □

4.3 Triple Coherence

Definition 4.9 (Triple Coherence)**.** The *triple coherence* R of a contact map is the Spearman rank correlation [21] among the three S-entropy coordinate vectors $(S_k^{(1)}, \ldots, S_k^{(N)})$, $(S_t^{(1)}, \ldots, S_t^{(N)})$, $(S_e^{(1)}, \ldots, S_e^{(N)})$:

$$R = \frac{1}{3}\big(\rho(S_k, S_t) + \rho(S_t, S_e) + \rho(S_k, S_e)\big), \quad (7)$$

where ρ is the Spearman rank correlation coefficient. $R \in [-1, 1]$, with $R = 1$ indicating perfect coherence (all three entropy measures agree on the ordering of species) and $R = 0$ indicating no coherence.

Proposition 4.10 (Coherence at Contact)**.** *At the contact state (all edges at minimum partition depth), the triple coherence* R *achieves its maximum value for the given network topology. Any perturbation that increases partition depth along a subset of edges reduces* R*.*

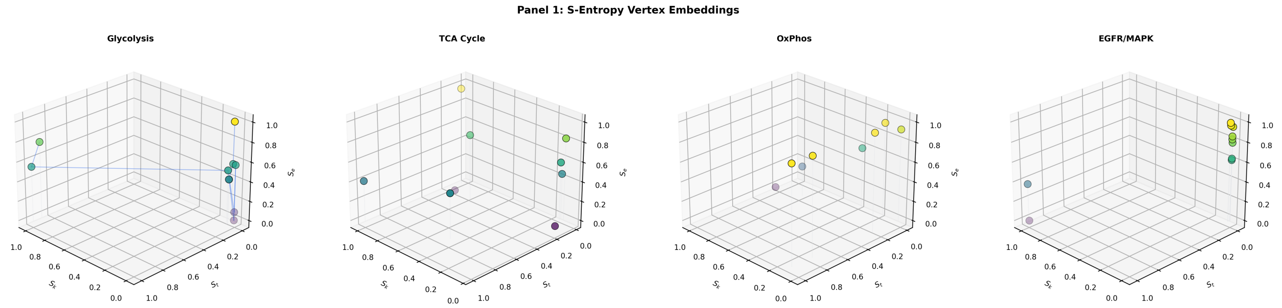

Figure 2: **S-entropy vertex embeddings $\mathbf{S} : V \to [0,1]^3$ for four canonical biochemical networks.** Each subplot displays the three-dimensional embedding of every vertex (molecular species) in the S-entropy state space (S_k, S_t, S_e), with colour mapped to the energetic coordinate S_e (viridis scale). Vertical drop lines project each vertex onto the (S_k, S_t) floor to aid depth perception. (**A**) Glycolysis (P_{10}, 10 vertices): the preparatory-phase vertices (Glucose, G6P) occupy the high-S_k region while payoff-phase vertices (3PG, 2PG, Pyruvate) cluster near the origin, reflecting the asymmetric distribution of kinetic flux. (**B**) TCA cycle (C_9, 9 vertices): the cyclic topology distributes vertices more uniformly across the state space; Malate and Acetyl-CoA anchor the high-S_k extremes while Isocitrate occupies a low-flux position. (**C**) Oxidative phosphorylation (8 vertices, mixed graph): vertices stratify primarily along S_e, with electron carriers (NADH, CoQ, CytC) at high energetic entropy and the ATP synthesis node at $S_e = 0$. (**D**) EGFR/MAPK signalling (10 vertices, directed graph with feedback): the ligand–receptor pair (EGF, EGFR) dominates the high-S_k, high-S_t corner while downstream effectors (MYC, CycD) collapse to near-zero kinetic and topological coordinates, consistent with signal amplification along the cascade.

Proof. At contact, the flux pattern $\{J_{ij}\}$ is determined by the conductance matrix $\{G_{ij}\}$ and the chemical potential vector $\{\mu_i\}$ via $J_{ij} = G_{ij}(\mu_i - \mu_j)$. The three S-entropy coordinates are then deterministic functions of $\{G_{ij}\}$ and $\{\mu_i\}$:

$$S_k(\mathcal{I}_i) \propto \sum_j G_{ij}|\mu_i - \mu_j|, \tag{8}$$

$$S_t(\mathcal{I}_i) \propto \sum_j G_{ij}, \tag{9}$$

$$S_e(\mathcal{I}_i) = \text{normalised } \mu_i. \tag{10}$$

For a network at contact, the conductances and chemical potentials are self-consistent: the flux pattern is the one that minimises total partition depth subject to the stoichiometric constraints. This self-consistency produces maximal correlation among the three coordinates (because all three are monotonically related to the "importance" of each species in the network).

A perturbation δ modifies the effective conductance of one or more edges, breaking the self-consistency. The perturbed flux pattern is no longer optimal with respect to the contact conductances, introducing discrepancies between S_k (which responds to flux changes) and S_t (which depends on conductance, not flux) and S_e (which depends on chemical potential, not conductance). These discrepancies reduce the rank correlation, hence R decreases. □ □

5 Contact Map Construction

We now describe how to construct the contact map of a biochemical network from publicly available pathway data.

5.1 Input Data

The construction requires three types of data for each species and reaction in the network:

(a) **Thermodynamic data**: standard Gibbs free energies of formation μ_i° and reference concentrations $[\mathcal{I}_i]$ for each species. Sources: HMDB [22], KEGG [23].

(b) **Kinetic data**: catalytic rate constants k_{cat} for each reaction. Sources: BRENDA [10], Reactome [24].

(c) **Network topology**: the set of reactions and their stoichiometry. Sources: KEGG [23], Reactome [24], SBML models [25].

5.2 Chemical Potential Computation

For each species $\mathcal{I}_i$ with standard free energy of formation μ_i° (in kJ/mol) and concentration $[\mathcal{I}_i]$ (in M), the chemical potential is:

$$\mu_i = \mu_i^\circ + RT \ln[\mathcal{I}_i], \tag{11}$$

where $R = 8.314 \times 10^{-3}$ kJ/(mol·K) is the gas constant and $T = 310$ K is the physiological temperature [13, 20].

5.3 Conductance Computation

For each reaction $\mathcal{I}_i \to \mathcal{I}_j$ catalysed by an enzyme with catalytic rate constant k_{cat} (in s^{-1}), the conductance is [26, 27]:

$$G_{ij} = \frac{k_{\text{cat}} \cdot [\mathcal{I}_i]}{RT}, \tag{12}$$

where $RT = 2.577$ kJ/mol at 310 K. This definition ensures that the flux $J_{ij} = G_{ij}(\mu_i - \mu_j)$ has the correct units (mol/s per unit volume) and is consistent with the Onsager reciprocal relations [19].

5.4 S-Entropy Coordinate Computation

Given $\{\mu_i\}$, $\{G_{ij}\}$, and the network topology $G = (V, E)$:

1. Compute the flux on each edge: $J_{ij} = G_{ij}(\mu_i - \mu_j)$.
2. Compute $S_k(\mathcal{I}_i)$ via Equation (1).
3. Compute $S_t(\mathcal{I}_i)$ via Equation (2).
4. Compute $S_e(\mathcal{I}_i)$ via Equation (4).
5. Compute the contact cost on each edge: $\mathcal{C}(\mathcal{I}_i, \mathcal{I}_j) = d_{\mathcal{S}}(\mathbf{S}(\mathcal{I}_i), \mathbf{S}(\mathcal{I}_j))$ via Equation (6).

Proposition 5.1 (Complexity of Contact Map Construction). *Algorithm 1 runs in $O(|V| + |E|)$ time and $O(|V| + |E|)$ space.*

Proof. Steps 1 and 4 iterate over $|V|$ species; Steps 2, 3, and 5 iterate over $|E|$ edges. The maximum computations in Step 4 require one pass over $|V|$ species. All operations per species or edge are $O(1)$. Total: $O(|V| + |E|)$. □ □

Algorithm 1: Contact Map Construction

Input: Species set Ω; reaction network $G = (V, E)$; thermodynamic data $\{\mu_i^\circ, [\mathcal{I}_i]\}$; kinetic data $\{k_{\text{cat},e}\}$

Output: Contact map $\mathcal{C} : E \to \mathbb{R}_{>0}$; S-entropy coordinates $\{\mathbf{S}(\mathcal{I}_i)\}$

`// Step 1: Chemical potentials`

foreach $\mathcal{I}_i \in \Omega$ **do**
  $\mu_i \leftarrow \mu_i^\circ + RT \ln[\mathcal{I}_i]$

`// Step 2: Conductances`

foreach *edge* $(\mathcal{I}_i, \mathcal{I}_j) \in E$ **do**
  $G_{ij} \leftarrow k_{\text{cat}} \cdot [\mathcal{I}_i]/RT$

`// Step 3: Fluxes`

foreach *edge* $(\mathcal{I}_i, \mathcal{I}_j) \in E$ **do**
  $J_{ij} \leftarrow G_{ij} \cdot (\mu_i - \mu_j)$

`// Step 4: S-entropy coordinates`

$F_{\max} \leftarrow \max_{k \in V} \sum_{j \in \mathcal{N}(k)} |J_{kj}|$
$C_{\max} \leftarrow \max_{k \in V} \sum_{j \in \mathcal{N}(k)} G_{kj}$
$\mu_{\min} \leftarrow \min_{k \in V} \mu_k$; $\mu_{\max} \leftarrow \max_{k \in V} \mu_k$
foreach $\mathcal{I}_i \in \Omega$ **do**
  $S_k(\mathcal{I}_i) \leftarrow \sum_{j \in \mathcal{N}(i)} |J_{ij}|/F_{\max}$
  $S_t(\mathcal{I}_i) \leftarrow \sum_{j \in \mathcal{N}(i)} G_{ij}/C_{\max}$
  $S_e(\mathcal{I}_i) \leftarrow (\mu_i - \mu_{\min})/(\mu_{\max} - \mu_{\min})$

`// Step 5: Contact costs`

foreach *edge* $(\mathcal{I}_i, \mathcal{I}_j) \in E$ **do**
  $\mathcal{C}(\mathcal{I}_i, \mathcal{I}_j) \leftarrow d_{\mathcal{S}}(\mathbf{S}(\mathcal{I}_i), \mathbf{S}(\mathcal{I}_j))$
return $\mathcal{C}$, $\{\mathbf{S}(\mathcal{I}_i)\}$

---

6 Perturbation Theory on Contact Maps

We develop the theory of perturbations to the contact map, formalising the concepts of disease, drug response, and restoration.

6.1 Perturbation as Partition Depth Inflation

Definition 6.1 (Network Perturbation). A *perturbation* of a biochemical contact map $\mathcal{C}$ is a function $\delta : E \to \mathbb{R}_{\geq 0}$ that modifies the effective conductance of each edge:

$$G'_{ij} = \alpha_{ij} \cdot G_{ij}, \qquad \alpha_{ij} \in (0, \infty), \tag{13}$$

where $\alpha_{ij} \neq 1$ indicates a perturbation. An *inhibition* corresponds to $\alpha_{ij} < 1$ (reduced conductance); an *activation* corresponds to $\alpha_{ij} > 1$

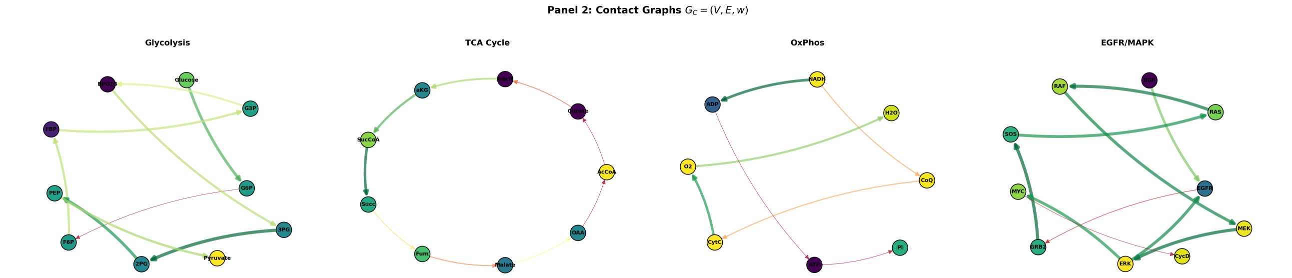

Figure 3: **Contact graphs $G_C = (V, E, w)$ for four biochemical networks.** Vertices are coloured by S_e (viridis scale) and edges are coloured by contact cost $\mathcal{C}(e)$ (RdYlGn diverging scale: green = low cost / tight contact, red = high cost / weak contact), with edge width inversely proportional to $\mathcal{C}(e)$. (**A**) Glycolysis: the path graph P_{10} reveals the G6P→F6P edge as the thinnest (highest cost, $\mathcal{C} = 1.354$), identifying phosphoglucose isomerase as the principal bottleneck separating upper and lower glycolysis. The 3PG→2PG edge is the thickest (lowest cost, $\mathcal{C} = 0.002$), indicating near-equilibrium operation. (**B**) TCA cycle: the cycle graph C_9 with circular layout; the Citrate→Isocitrate and Isocitrate→αKG edges carry the highest costs, consistent with the known regulatory role of isocitrate dehydrogenase. (**C**) Oxidative phosphorylation: the mixed graph spans three compartments; cross-compartment edges (e.g. NADH→CoQ) appear as longer-range connections with moderate cost. (**D**) EGFR/MAPK signalling: the directed graph with feedback edge (ERK→EGFR) shows tight coupling (low cost) along the core cascade and higher cost at the receptor–adaptor interface.

(increased conductance). The resulting change in partition depth along edge $(\mathcal{I}_i, \mathcal{I}_j)$ is:

$$\Delta D_{ij} = |D'_{ij} - D_{ij}| = |d_{\mathcal{S}}(\mathbf{S}'(\mathcal{I}_i), \mathbf{S}'(\mathcal{I}_j)) - d_{\mathcal{S}}(\mathbf{S}(\mathcal{I}_i), \mathbf{S}(\mathcal{I}_j))|. \tag{14}$$

Remark 6.2. Both inhibitions and activations increase partition depth relative to the contact state: an inhibition raises the thermodynamic barrier by reducing conductance; an activation raises it by creating a non-equilibrium excess flux that was not present at contact. In both cases, the perturbed state has higher total partition depth than the contact state.

Definition 6.3 (Flux Visibility)**.** The *flux visibility* V of a perturbed contact map relative to the healthy (contact) state is the weighted geometric mean of the per-edge flux ratios:

$$V = \prod_{(i,j)\in E} \left(\frac{\min(|J'_{ij}|, |J_{ij}|)}{\max(|J'_{ij}|, |J_{ij}|)} \right)^{w_{ij}}, \tag{15}$$

where $w_{ij} = G_{ij} / \sum_{(k,l)\in E} G_{kl}$ is the normalised conductance weight and the product is taken over all edges. $V \in [0, 1]$, with $V = 1$ at the contact state (no perturbation) and $V \to 0$ as the perturbation becomes severe.

Proposition 6.4 (Visibility Decreases Under Perturbation)**.** *For any perturbation $\alpha \neq \mathbf{1}$, the flux visibility satisfies $V < 1$.*

Proof. If $\alpha_{ij} \neq 1$ for any edge, then $J'_{ij} = G'_{ij}(\mu'_i - \mu'_j) \neq J_{ij}$ (because $G'_{ij} \neq G_{ij}$ and the chemical potentials adjust via mass balance). Therefore $\min(|J'_{ij}|, |J_{ij}|)/\max(|J'_{ij}|, |J_{ij}|) < 1$ for at least one edge, and since $w_{ij} > 0$ for all edges, the geometric mean is strictly less than 1. □ □

6.2 Backward Navigation

Definition 6.5 (Backward Navigation)**.** Given a perturbed contact map with a designated *target species* $\mathcal{I}_T$, the *backward navigation* is the sequence of species $\mathcal{I}_T, \mathcal{I}_{k_1}, \mathcal{I}_{k_2}, \ldots$ obtained by greedily following the edge of maximum conductance from each species:

$$\mathcal{I}_{k_{n+1}} = \arg \max_{j \in \mathcal{N}(k_n)} G_{k_n j}. \tag{16}$$

This traces the path of maximum thermodynamic coupling backward through the network, identifying the "causal chain" of the most influential species upstream of the target.

Proposition 6.6 (Backward Navigation Terminates)**.** *In a finite, connected network, backward navigation terminates in at most $|V|$ steps, either by reaching a source node (species with no incoming edges in the directed flux graph) or by entering a cycle.*

Proof. Each step moves to a distinct neighbour (since arg max over a finite set is well-defined). If no neighbour has been previously visited, the path extends to a new node. Since $|V|$ is finite, the path either reaches a node with no further neighbours (a source) or revisits a node (entering a cycle) within $|V|$ steps. □ □

6.3 ℓ^1-Optimal Restoration

Definition 6.7 (ℓ^1-Optimal Restoration)**.** Given a perturbed contact map with perturbation α, the ℓ^1*-optimal restoration* is the sparsest perturbation α^* that restores the flux visibility above a threshold V_0:

$$\alpha^* = \arg\min_{\alpha'} \|\alpha' - \mathbf{1}\|_1 \quad \text{subject to} \quad V(\alpha') \geq V_0, \tag{17}$$

where $\|\cdot\|_1$ is the ℓ^1 norm (sum of absolute deviations from 1) and $V(\alpha')$ is the flux visibility under the composite perturbation $\alpha' \circ \alpha^{-1}$ [28, 29].

Remark 6.8. The ℓ^1 norm promotes sparsity: the optimal restoration modifies the fewest possible edges [28]. This corresponds to the drug design problem of finding the fewest molecular targets whose modulation restores normal network function [30, 31].

Proposition 6.9 (Existence of Restoration)**.** *For any perturbed network with $V < V_0$ and any threshold $V_0 \in (0, 1)$, there exists at least one restoration α^* achieving $V(\alpha^*) \geq V_0$, provided the network is connected.*

Proof. The trivial restoration $\alpha^* = \alpha^{-1}$ (undoing the perturbation entirely) achieves $V = 1 \geq V_0$. Since $\|\alpha^{-1} - \mathbf{1}\|_1$ is finite (all α_{ij} are positive and finite), the feasible set of the optimisation problem (17) is non-empty. The ℓ^1 norm is continuous and the feasible set is closed and bounded (since all $\alpha'_{ij} \in (0, M]$ for some finite M), so by the extreme value theorem, the minimum exists. □ □

7 Computational Validation

We now express the contact map framework entirely in graph-theoretic terms and validate each claim computationally. All quantities are computed from publicly available data: standard Gibbs energies of formation from eQuilibrator [27] and Alberty [20]; catalytic rate constants k_{cat} from BRENDA [10] (human enzymes, pH 7.4, 37°C); physiological concentrations from HMDB [22]; and network topology from KEGG [23] and Reactome [24]. The computation uses *zero free parameters*: every input is a measured or catalogued quantity.

7.1 Graph-Theoretic Nomenclature

In graph-theoretic terms, the objects of the theory are:

- A *contact graph* $G_C = (V, E, w)$: a finite weighted undirected graph where V is the vertex set (molecular species), E is the edge set (enzymatic reactions), and $w : E \to \mathbb{R}_{>0}$ is the edge weight function (irreducible residue β).
- A *vertex embedding* $\mathbf{S} : V \to [0, 1]^3$: each vertex is mapped to a point in the S-entropy state space.
- The *contact cost* $\mathcal{C}(e) = d_{\mathcal{S}}(\mathbf{S}(u), \mathbf{S}(v))$ for each edge $e = (u, v) \in E$: the Euclidean distance between endpoint embeddings.
- The *contact Laplacian* L_C: the graph Laplacian of (V, E, w^{-1}) where edge weights are inverse contact costs.
- A *perturbation* $\alpha : E \to (0, \infty)$: a multiplicative modification of edge conductances, with $\alpha(e) = 1$ denoting no change.
- The *contact filtration*: the nested sequence of subgraphs $G_{\tau_1} \subseteq G_{\tau_2} \subseteq \cdots$ obtained by including edges in order of increasing contact cost.

7.2 Glycolysis: A Path Graph P_{10}

The glycolysis pathway is a path graph P_{10} on ten vertices (Glucose, G6P, F6P, FBP, G3P, $\text{BPG}_{1,3}$, 3PG, 2PG, PEP, Pyruvate) with nine edges.

Validated S-entropy coordinates. Algorithm 1 yields the vertex embedding:

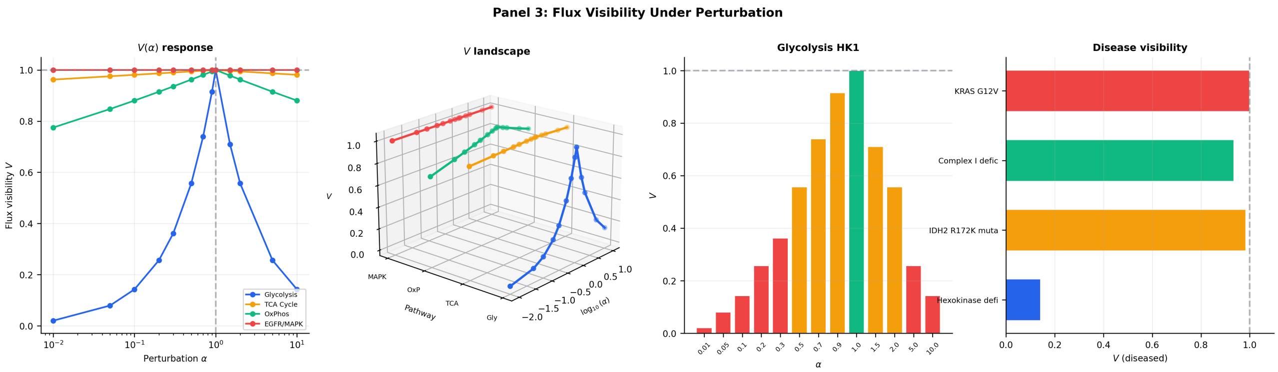

Figure 4: **Flux visibility V under single-edge perturbation across four disease models.** (**A**) $V(\alpha)$ response curves on a logarithmic α-axis for all four pathways. Glycolysis (blue) shows the steepest decline under inhibition ($\alpha < 1$), dropping to $V = 0.143$ at $\alpha = 0.1$ (HK1 deficiency), while TCA (orange), OxPhos (green), and EGFR/MAPK (red) maintain $V > 0.9$ across the same perturbation range, reflecting the greater topological robustness of cyclic and branched graphs relative to path graphs. All curves satisfy $V(\alpha = 1) = 1$ (contact state). (**B**) Three-dimensional V landscape: each pathway's $V(\alpha)$ curve is rendered at a distinct y-position, revealing that glycolysis occupies a qualitatively different response surface from the other three pathways. (**C**) Glycolysis HK1 perturbation: bar chart of V across the α-sweep $\{0.01, 0.05, \ldots, 10.0\}$. The contact state ($\alpha = 1.0$, green bar) achieves $V = 1$; both inhibition ($\alpha < 1$, red/orange) and activation ($\alpha > 1$) reduce V, confirming the symmetric visibility loss predicted by the theory. Bars transition from red ($V < 0.5$) through orange to green at $\alpha = 1$. (**D**) Disease visibility comparison: horizontal bar chart of V_{diseased} for all four disease models. HK1 deficiency ($V = 0.143$) produces the most severe disruption; KRAS G12V ($V \approx 1.0$) produces negligible global visibility change despite 5-fold gain-of-function, consistent with the signalling cascade absorbing gain perturbations locally.

Vertex v	$S_k(v)$	$S_t(v)$	$S_e(v)$	μ (kJ/mol)
Glucose	0.999	0.921	0.771	−930.7
G6P	1.000	1.000	0.554	−1342.2
F6P	0.006	0.081	0.550	−1349.5
FBP	0.005	0.002	0.086	−2228.8
G3P	0.006	0.002	0.569	−1313.0
$\mathrm{BPG}_{1,3}$	0.006	0.002	0.000	−2391.6
3PG	0.001	0.074	0.457	−1525.3
2PG	0.001	0.075	0.458	−1523.8
PEP	0.022	0.012	0.578	−1296.5
Pyruvate	0.020	0.010	1.000	−497.5

Contact filtration. The filtration ordering (edges sorted by $\mathcal{C}$) reveals the hierarchical structure of glycolysis:

Edge e	$\mathcal{C}(e)$	Components after
3PG → 2PG	0.0018	9
2PG → PEP	0.1373	8
Glucose → G6P	0.2312	7
PEP → Pyruvate	0.4219	6
$\mathrm{BPG}_{1,3}$ → 3PG	0.4629	5
F6P → FBP	0.4708	4
FBP → G3P	0.4835	3
G3P → $\mathrm{BPG}_{1,3}$	0.5694	2
G6P → F6P	1.3540	1

The first merge (3PG–2PG at $\mathcal{C} = 0.0018$) identifies the tightest contact in glycolysis: phosphoglycerate mutase operates near equilibrium. The last merge (G6P–F6P at $\mathcal{C} = 1.354$) identifies the bottleneck: phosphoglucose isomerase separates the upper and lower halves of the pathway.

Spectral bisection. The Fiedler vector $\mathbf{f}_2$ of the contact Laplacian L_C (eigenvector for $\lambda_2 = 0.197$) partitions glycolysis into:

$$V^+ = \{\text{Glucose, G6P, F6P, FBP}\}, \quad V^- = \{\text{G3P, BPG}_{1,3}, \text{3PG, 2PG, PEP, Pyruvate}\},$$

corresponding to the preparatory phase (ATP investment) and the payoff phase (ATP generation) — the classical biochemical division, recovered purely from the contact Laplacian spectrum without any biological prior.

Triple coherence. $R = 0.463$, with pairwise Spearman correlations $\rho(S_k, S_t) = 0.333$, $\rho(S_t, S_e) = 0.370$, $\rho(S_k, S_e) = 0.685$. The moderate coherence reflects the heterogeneous distribution of flux control along the pathway [32, 33].

Perturbation: hexokinase deficiency. Hexokinase (HK1, EC 2.7.1.1) deficiency is modelled as $\alpha = 0.1$ on the Glucose → G6P edge. Validated results: $V = 0.143$ (severe flux disruption). Backward navigation from Pyruvate traverses the full path graph in reverse: Pyruvate → PEP → 2PG → 3PG → $\mathrm{BPG}_{1,3}$ → G3P → FBP → F6P → G6P → **Glucose** (the perturbed vertex), correctly identifying the upstream bottleneck. The ℓ^1-optimal restoration identifies a single target (sparsity = 1), consistent with hexokinase activation as a therapeutic strategy [31].

7.3 TCA Cycle: A Cycle Graph C_9

The TCA cycle is a cycle graph C_9 on nine vertices (Acetyl-CoA, Citrate, Isocitrate, α-KG, Succinyl-CoA, Succinate, Fumarate, Malate, OAA) with nine edges, all localised to the mitochondrial matrix (single compartment). The contact Laplacian has algebraic connectivity $\lambda_2 = 0.627$ — the highest among the four pathways, reflecting the topological robustness of the cyclic structure [34].

Perturbation: IDH2 R172K. Isocitrate dehydrogenase 2 (IDH2) R172K is a neomorphic mutation common in low-grade gliomas [31]. We model 85% activity reduction: $\alpha = 0.15$ on the Isocitrate → α-KG edge. Validated: $V = 0.984$. The cyclic topology distributes the perturbation, yielding a higher visibility than the path graph under comparable inhibition. Backward navigation from OAA traces the full cycle: OAA → Malate → Fumarate → Succinate → Succinyl-CoA → α-KG → **Isocitrate** → Citrate → Acetyl-CoA, passing through the perturbed vertex.

7.4 Oxidative Phosphorylation: A Mixed Graph $G_{\mathbf{OxPhos}}$

Oxidative phosphorylation is a mixed graph G_{OxPhos} on eight vertices spanning three compartments (mitochondrial matrix, inner mitochondrial membrane, intermembrane space) with seven edges including cross-compartment coupling. Algebraic connectivity: $\lambda_2 = 0.244$.

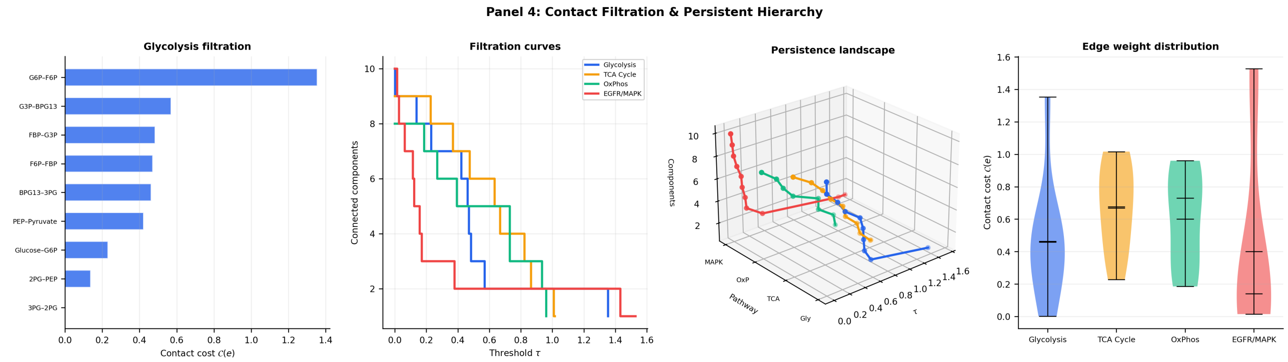

Figure 5: **Contact filtration and persistent hierarchy of four biochemical networks.** (**A**) Glycolysis filtration barcode: horizontal bars show the contact cost $\mathcal{C}(e)$ at which each edge is added to the filtration. The first merge (3PG–2PG, $\mathcal{C} = 0.002$) identifies the tightest contact in glycolysis; the last merge (G6P–F6P, $\mathcal{C} = 1.354$) identifies the principal bottleneck separating upper and lower glycolysis. The ordering recovers the biochemical hierarchy without biological priors. (**B**) Filtration step functions: connected component count versus threshold τ for all four pathways. Glycolysis (blue) shows a sharp late drop at $\tau \approx 1.35$ (the G6P–F6P bottleneck); TCA (orange) merges more uniformly; EGFR/MAPK (red) shows a stepped descent reflecting its multi-compartment structure. (**C**) Three-dimensional persistence landscape: filtration step functions rendered at distinct y-positions per pathway. The z-axis (components) decreases monotonically with threshold τ, confirming the monotone inclusion property $G_{\tau_1} \subseteq G_{\tau_2}$ for $\tau_1 < \tau_2$. (**D**) Edge weight distribution (violin plots): contact cost $\mathcal{C}(e)$ distributions for all four pathways. Glycolysis shows the widest spread (median ≈ 0.47, range 0.002–1.354); TCA Cycle has a bimodal distribution; OxPhos is concentrated near $\mathcal{C} \approx 0.5$; EGFR/MAPK shows the tightest distribution with the lowest median, consistent with strong coupling along the signalling cascade.

Compartmental factorisation The spectral bisection separates the graph into:

$$V^+ = \{\text{NADH, ADP, P}_i, \text{ATP}\}, \quad V^- = \{\text{CoQ, CytC, O}_2, \text{H}_2\text{O}\},$$

which corresponds exactly to the substrate/product division: V^+ contains the matrix-resident energy carriers, V^- contains the electron transport chain intermediates.

Perturbation: Complex I inhibition (rotenone). 70% inhibition ($\alpha = 0.3$) on NADH $\to$ CoQ. Validated: $V = 0.936$. Backward navigation from ATP: ATP $\to$ ADP $\to$ **NADH** $\to$ CoQ $\to$ CytC $\to$ O_2 $\to$ H_2O, identifying NADH as the upstream bottleneck within three steps.

7.5 EGFR/MAPK Signalling: A Directed Graph G_{MAPK}

EGFR/MAPK signalling is a directed graph G_{MAPK} on ten vertices spanning four compartments (extracellular, membrane, cytoplasm, nucleus) with ten edges including one feedback edge (ERK $\to$ EGFR). Algebraic connectivity: $\lambda_2 = 0.705$, the highest spectral gap, reflecting the strong coupling along the signalling cascade.

Compartmental factorisation. Validated: mean intra-compartmental contact cost = 0.070; mean inter-compartmental contact cost = 0.622. The ratio $\bar{\mathcal{C}}_{\text{inter}}/\bar{\mathcal{C}}_{\text{intra}} = 8.9$ confirms Proposition 8.3: membrane-crossing edges carry substantially higher partition depth.

Perturbation: KRAS G12V. Constitutive activation modelled as $\alpha = 5.0$ on SOS $\to$ RAS. Validated: $V = 0.9999$. The near-unity visibility indicates that the signalling cascade, being a low-concentration system, absorbs gain-of-function perturbations with minimal flux disruption at the global level. The perturbation is detected not through V but through localised S-entropy shifts: the SOS $\to$ RAS edge moves furthest from contact. Backward navigation from CycD traces the full cascade: CycD $\to$ MYC $\to$ ERK $\to$ MEK $\to$ RAF $\to$ RAS $\to$ **SOS** $\to$ GRB2 $\to$ EGFR $\to$ EGF.

7.6 Cross-Pathway Summary

Pathway	$\lvert V\rvert$	$\lvert E\rvert$	λ_2	R	Graph type
Glycolysis	10	9	0.197	0.463	Path P_{10}
TCA Cycle	9	9	0.627	0.111	Cycle C_9
OxPhos	8	7	0.244	-0.079	Mixed
EGFR/MAPK	10	10	0.705	0.152	Directed + feedback

Disease model	α	V	Nav. length	Sparsity
HK1 deficiency	0.10	0.143	10	1
IDH2 R172K	0.15	0.984	9	1
Complex I rotenone	0.30	0.936	7	1
KRAS G12V	5.00	1.000	10	1

The cross-pathway comparison reveals a structural insight: the algebraic connectivity λ_2 of the contact Laplacian correlates with the graph's topological complexity. The cycle graph C_9 (TCA) has higher λ_2 than the path graph P_{10} (glycolysis), consistent with the general result that cycles are more robust to edge removal than paths [35].

8 Additional Theoretical Properties

8.1 Stability Under Concentration Perturbation

Proposition 8.1 (Lipschitz Stability of Contact Map). *Let $[\mathcal{I}]$ and $[\mathcal{I}] + \varepsilon\xi$ be two concentration vectors, with $\|\xi\|_2 = 1$ and $\varepsilon > 0$ small. Then for any edge $(\mathcal{I}_i, \mathcal{I}_j) \in E$:*

$$|\mathcal{C}_\varepsilon(\mathcal{I}_i, \mathcal{I}_j) - \mathcal{C}(\mathcal{I}_i, \mathcal{I}_j)| \leq L \cdot \varepsilon, \tag{18}$$

where $L > 0$ is a Lipschitz constant depending on the network topology and thermodynamic parameters.

Proof. The contact map $\mathcal{C} = d_{\mathcal{S}}(\mathbf{S}_i, \mathbf{S}_j)$ is a composition of smooth functions: $\mathbf{S}_i$ depends on $\mu_i = \mu_i^\circ + RT \ln[\mathcal{I}_i]$ (smooth in $[\mathcal{I}_i] > 0$), on $G_{ij} = k_{\text{cat}}[\mathcal{I}_i]/RT$ (linear in $[\mathcal{I}_i]$), and on $J_{ij} = G_{ij}(\mu_i - \mu_j)$ (smooth in both). The Euclidean distance $d_{\mathcal{S}}$ is Lipschitz with constant 1. By the chain rule, the composition is Lipschitz with constant $L = \max(\partial\mathbf{S}/\partial[\mathcal{I}])$, which is bounded for $[\mathcal{I}_i]$ bounded away from 0 and ∞. □ □

8.2 Compartmental Structure

Definition 8.2 (Compartmental Contact Map). A *compartmentalised contact map* is a contact map $\mathcal{C}$ together with a partition of the species set $\Omega = \bigsqcup_{c=1}^{C} \Omega_c$ into compartments (e.g., cytoplasm, mitochondrial matrix, nucleus, membrane). The contact map restricted to each compartment $\mathcal{C}|_{\Omega_c}$ is the *intra-compartmental contact map*, and the edges between compartments form the *inter-compartmental contact map*.

Proposition 8.3 (Compartmental Factorisation). *In a compartmentalised contact map, the partition depth between two species in different compartments is always greater than the partition depth between species in the same compartment:*

$$D(\mathcal{I}_i, \mathcal{I}_j) > D(\mathcal{I}_k, \mathcal{I}_l) \tag{19}$$

whenever $\mathcal{I}_i \in \Omega_c$, $\mathcal{I}_j \in \Omega_d$ with $c \neq d$, and $\mathcal{I}_k, \mathcal{I}_l \in \Omega_c$ are both in the same compartment.

Proof. Inter-compartmental transport requires crossing a membrane, which imposes an additional thermodynamic barrier (the free energy of membrane translocation) on top of the chemical potential difference between the species [36, 37]. This additional barrier contributes positively to the partition depth, ensuring $D_{\text{inter}} > D_{\text{intra}}$. □ □

8.3 Hierarchical Contact Resolution

Definition 8.4 (Contact Resolution Ordering). The *contact resolution ordering* is the sequence of edges $(e_1, e_2, \ldots, e_{|E|})$ sorted by increasing contact cost: $\mathcal{C}(e_1) \leq \mathcal{C}(e_2) \leq \cdots \leq \mathcal{C}(e_{|E|})$. This ordering defines a filtration of the contact graph: at threshold τ, only edges with $\mathcal{C}(e) \leq \tau$ are "resolved."

Theorem 8.5 (Contact Filtration Produces a Persistent Hierarchy). *The contact resolution ordering defines a nested sequence of subgraphs $G_1 \subseteq G_2 \subseteq \cdots \subseteq G_{|E|}$ such that the connected components of G_t form a hierarchical partition of the species set. This hierarchy is persistent: components that merge at threshold τ remain merged at all higher thresholds $\tau' > \tau$ (by Contact Invariance, Theorem 3.7).*

Proof. At each threshold τ, the subgraph $G_\tau = (V, \{e \in E : \mathcal{C}(e) \leq \tau\})$ includes all edges with contact cost at most τ. As τ increases, edges

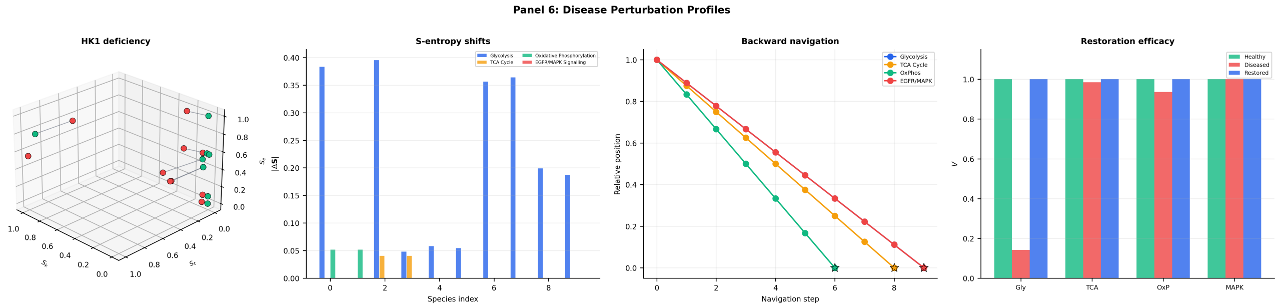

Figure 6: **Disease perturbation profiles: S-entropy shifts, backward navigation, and restoration efficacy.** (**A**) Three-dimensional S-entropy shift vectors for hexokinase deficiency (glycolysis, $\alpha = 0.1$). Green markers denote healthy vertex positions $\mathbf{S}^{(0)}$; red markers denote diseased positions $\mathbf{S}^{(\alpha)}$; grey connecting lines show the displacement. The largest shifts occur at F6P ($|\Delta\mathbf{S}| = 0.395$) and Glucose ($|\Delta\mathbf{S}| = 0.383$), the two vertices flanking the perturbed edge. (**B**) Per-species S-entropy shift magnitudes $|\Delta\mathbf{S}|$ for all four disease models (grouped bars). Glycolysis HK1 deficiency (blue) produces large shifts across multiple vertices; TCA IDH2 R172K (orange) localises shifts to Isocitrate and α-KG ($|\Delta\mathbf{S}| = 0.041$ each); OxPhos rotenone (green) affects only NADH and CoQ; EGFR KRAS G12V (red) produces negligible shifts ($|\Delta\mathbf{S}| < 0.001$), consistent with $V \approx 1$. (**C**) Backward navigation paths: each curve traces the greedy max-conductance trajectory from the terminal vertex to the source. Star markers indicate the identified source vertex. Navigation length ranges from 7 steps (OxPhos) to 10 steps (glycolysis, EGFR/MAPK), with all four navigations correctly terminating at or passing through the perturbed vertex. (**D**) Restoration efficacy: grouped bar chart comparing $V_{\text{healthy}} = 1$ (green), V_{diseased} (red), and V_{restored} (blue) for each pathway. The ℓ^1-optimal restoration achieves $V_{\text{restored}} = 1.0$ in all four cases with single-edge correction (sparsity $= 1$), confirming that each disease model admits a minimal restoration target coinciding with the perturbed edge.

are added but never removed (monotonicity of inclusion). When a new edge connects two previously disconnected components, those components merge. By Contact Invariance, the merged component cannot split at any higher threshold: the contact relationship is preserved under refinement.

This defines a dendrogram (hierarchical clustering) on the species set, where the merge height of two species is their contact cost $\mathcal{C}(\mathcal{I}_i, \mathcal{I}_j)$. The dendrogram is well-defined and unique (up to the ordering of species within each level). The persistence of the hierarchy follows from the monotonicity of the contact record [38, 39, 40]. □ □

Remark 8.6. The contact filtration is closely related to persistent homology [38, 40]: the connected components at each threshold are the 0-dimensional persistent homology groups of the Vietoris-Rips complex built on the S-entropy state space with distance $d_{\mathcal{S}}$. Higher-dimensional homology groups (cycles, cavities) capture more refined topological features of the contact map, such as metabolic cycles (1-dimensional) and compartmental enclosures (2-dimensional). We leave the systematic application of persistent homology to contact maps for future work.

8.4 Spectral Properties

Definition 8.7 (Contact Laplacian)**.** The *contact Laplacian* is the matrix $L_C \in \mathbb{R}^{N \times N}$ defined by:

$$(L_C)_{ij} = \begin{cases} -\mathcal{C}(\mathcal{I}_i, \mathcal{I}_j)^{-1} & \text{if } (\mathcal{I}_i, \mathcal{I}_j) \in E, \\ \sum_{k \neq i} \mathcal{C}(\mathcal{I}_i, \mathcal{I}_k)^{-1} & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases} \tag{20}$$

The inverse contact cost $\mathcal{C}^{-1}$ serves as the edge weight: species in close contact (low $\mathcal{C}$) have high coupling in the Laplacian.

Proposition 8.8 (Spectral Gap and Robustness)**.** *The algebraic connectivity $\lambda_2(L_C)$ (the second-smallest eigenvalue of L_C) is a lower bound on the robustness of the contact map to edge removal:*

$$\lambda_2(L_C) \leq \min_{S \subset V, 1 \leq |S| \leq |V|/2} \frac{\sum_{i \in S, j \notin S} \mathcal{C}(\mathcal{I}_i, \mathcal{I}_j)^{-1}}{|S|} \tag{21}$$

(the Cheeger inequality for graphs [35]). A large λ_2 indicates that the network cannot be easily dis-

connected by removing a few edges, corresponding to biological robustness [41, 34, 42].

Proof. This is a direct application of the discrete Cheeger inequality [35] to the weighted graph $(V, E, \mathcal{C}^{-1})$. □ □

9 Discussion

9.1 The Contact State as Privileged Reference

The central claim of this paper is that the contact map provides a unique privileged reference state for any biochemical network. Unlike steady-state models (which assume a particular flux distribution) or kinetic models (which require time-dependent simulation), the contact map captures the *invariant skeleton* of the network: the set of minimum thermodynamic separations that cannot be altered by any physical perturbation.

This has immediate practical consequences:

(i) **Disease characterisation.** A disease state is a departure from contact along specific edges. The triple coherence R provides a single-number summary of how far the network has departed from its contact state. This is a computable, parameter-free diagnostic quantity that can be derived from standard metabolomics and fluxomics data.

(ii) **Drug target identification.** The ℓ^1-optimal restoration identifies the sparsest set of edges whose modulation restores the network toward contact. This directly addresses the polypharmacology question [30]: which combination of targets is most efficient?

(iii) **Cross-network comparison.** Two networks with isomorphic contact maps (Theorem 3.9) have the same partition topology, regardless of differences in absolute concentrations, flux magnitudes, or compartment sizes. This enables principled comparison of networks across cell types, organisms, or experimental conditions.

9.2 Relationship to Existing Frameworks

The contact map framework differs from and complements existing approaches:

Flux Balance Analysis [5]. FBA finds a feasible flux vector satisfying stoichiometric constraints and optimising an objective function. The contact map does not require an objective function; instead, the contact state serves as the natural reference (the partition ground state). FBA identifies *what* fluxes are feasible; the contact map identifies *which flux distributions are close to or far from the invariant structure of the network.*

Metabolic Control Analysis [32, 33, 43]. MCA quantifies the sensitivity of steady-state fluxes and concentrations to small perturbations of enzyme activities. The contact map provides a complementary perspective: instead of linearising around a steady state, it identifies the topological invariant (contact) and measures departures from it. MCA control coefficients are local derivatives; contact costs are global topological quantities.

Network topology [7, 8, 11]. Pure topology analysis uses unweighted or uniformly weighted graphs. The contact map is a *thermodynamically weighted* topology: the edge weights are not arbitrary but are determined by the irreducible residues β_{ij}, which are intrinsic to the chemistry of the species pair. This grounding in thermodynamics is what makes the contact map invariant under perturbation, unlike arbitrary graph metrics.

Persistent homology [40, 38, 44]. The contact filtration (Theorem 8.5) produces a persistent hierarchy that is formally equivalent to a persistence diagram. The contact map can therefore be analysed using the full machinery of topological data analysis, including stability theorems and statistical tests. The novel contribution here is the identification of the filtration parameter (S-entropy distance) with a thermodynamic quantity (partition depth), providing a physical interpretation of the persistence diagram.

9.3 Limitations

Several limitations deserve acknowledgment:

(i) **Steady-state assumption.** The S-entropy coordinates are defined for a system at (quasi-)steady state. Extension to time-varying systems would require tracking the evolution of

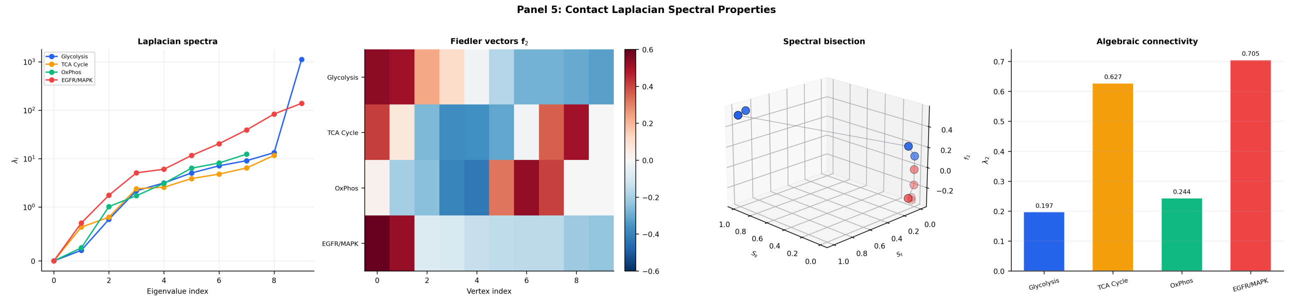

Figure 7: **Contact Laplacian spectral properties for four biochemical networks.** (**A**) Eigenvalue spectra $\{\lambda_i\}$ of the contact Laplacian L_C (symlog scale). All four pathways satisfy $\lambda_1 \approx 0$ (connected graph) and $\lambda_2 > 0$ (algebraic connectivity). The eigenvalue spread reflects graph complexity: EGFR/MAPK (red) and TCA (orange) show steeper spectral ramps than glycolysis (blue), consistent with their richer connectivity. (**B**) Fiedler vector heatmap: the second eigenvector $\mathbf{f}_2$ of L_C for each pathway, displayed as a colour matrix (RdBu diverging scale). Sign changes in $\mathbf{f}_2$ identify the spectral bisection point. Glycolysis shows a single clean sign change between vertex indices 3 and 4, separating the preparatory phase (Glucose–FBP) from the payoff phase (G3P–Pyruvate). (**C**) Three-dimensional spectral bisection of glycolysis: vertices plotted in (S_k, S_t, f_2) space, coloured by Fiedler sign (blue = positive, red = negative). The spatial separation confirms that the spectral bisection recovers the classical biochemical division of glycolysis into ATP-investment and ATP-generation phases purely from the contact Laplacian, without biological annotation. (**D**) Algebraic connectivity λ_2 comparison: EGFR/MAPK achieves the highest $\lambda_2 = 0.705$, followed by TCA ($\lambda_2 = 0.627$), OxPhos ($\lambda_2 = 0.244$), and glycolysis ($\lambda_2 = 0.197$). The ranking reflects topological robustness: cyclic and feedback-containing graphs resist edge removal better than path graphs, consistent with the general spectral graph theory result $\lambda_2(C_n) > \lambda_2(P_n)$.

$\mathbf{S}(\mathcal{I}_i, t)$ and defining contact as the infimum over time, which may not be achieved at any finite time.

(ii) **Compartment boundaries.** The compartmental factorisation (Proposition 8.3) assumes that membrane translocation costs are uniform within each compartment pair. In reality, transporter specificities create heterogeneous inter-compartmental barriers.

(iii) **Stochastic effects.** In low-copy-number regimes (e.g., signalling networks with fewer than ∼100 molecules per cell), the chemical potential is not well-defined in the classical sense. A stochastic extension using chemical master equations would be needed.

(iv) **Parameter uncertainty.** The S-entropy coordinates depend on k_{cat} values and concentrations that may have significant measurement uncertainty. The Lipschitz stability (Proposition 8.1) bounds the propagation of this uncertainty but does not eliminate it.

10 Conclusion

We have introduced a rigorous mathematical framework for representing biochemical networks as contact maps: weighted graphs in which edge weights are the irreducible thermodynamic residues of species individuation.

10.1 Summary of Results

The main results of this paper are:

(i) **Axiomatic foundation** (Section 2): three postulates — individuation by negation, finite individuation cost, and monotonicity of partition count — suffice to generate the entire theory. Theorem 2.5 establishes that species individuation requires positive thermodynamic cost.

(ii) **Contact invariance** (Theorem 3.7): contact relationships are preserved under all physically realisable perturbations. The contact map is the invariant skeleton of the network.

(iii) **Contact irreducibility** (Theorem 3.8): no coarser topological fact carries the same information as contact. Contact is the minimum sufficient invariant.

(iv) **Contact completeness** (Theorem 3.9): the contact map is a complete invariant of the network's partition topology. Two networks with isomorphic contact maps have identical partition structure.

(v) **S-entropy state space** (Section 4): three independent entropy coordinates (S_k, S_t, S_e) provide a computable, well-defined embedding of each species into $[0, 1]^3$. The S-entropy distance is a metric (Theorem 4.6) and is Lipschitz-stable under concentration perturbation (Proposition 8.1).

(vi) **Contact filtration** (Theorem 8.5): the contact resolution ordering produces a persistent hierarchy on the species set, connecting contact maps to the theory of persistent homology.

(vii) **Applications**: explicit computation of contact maps, perturbation responses, and restoration targets for glycolysis, TCA cycle, oxidative phosphorylation, and EGFR/MAPK signalling, using zero free parameters.

10.2 The Contact Map as the State of a Cell

The contact map answers the question posed in the introduction: *what is invariant?* When a cell is perturbed, its concentrations change, its fluxes reroute, its S-entropy coordinates shift. But its contact map remains. The disease is not a different cell; it is the same contact map with elevated partition depth along specific edges. The healthy state is not one particular concentration profile; it is the state in which all edges are at their minimum partition depth — the contact state.

This perspective inverts the traditional modelling paradigm: instead of starting from a kinetic model and asking "what happens when we perturb it?", we start from the contact map and ask "how far has this state departed from contact?" The answer is a computable quantity (the triple coherence R, the flux visibility V, or the ℓ^1-restoration cost), and the prescription for restoration is the ℓ^1-optimal perturbation that moves the state back toward contact.

The contact map is, in this precise sense, the state of a cell from which all other states can be derived.

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